



University of Kent

End-of-term assignment

[Individual Written Assessment on
Mathematical Modelling and Computation]

Course code: BUSB7030

Name: Ali Nuriev

ID: 26006333

Department: Kent Business School

[Submission date: 3rd of June 2026]

Contents

Question 1: Optimally Assigning Crews to Flights	3
1. <i>Network Representation</i>	<i>3</i>
2. <i>Mathematical Formulation.....</i>	<i>4</i>
3. <i>Model Implementation</i>	<i>5</i>
4. <i>Results and discussion.....</i>	<i>6</i>
Question 2: COVID-19 Healthcare Facility Capacity Optimization Problem	6
1. <i>Mathematical formulation</i>	<i>6</i>
2&3. <i>Coding and presenting results.....</i>	<i>8</i>
4. <i>Sensitivity analysis</i>	<i>9</i>
Appendix	11
Question 1: Optimally Assigning Crews to Flights	11
Question 2: COVID-19 Healthcare Facility Capacity Optimization Problem.....	11

Question 1: Optimally Assigning Crews to Flights

1. Network Representation

Omega Airlines wants to determine the optimal way of assigning flight crews to the following different flights.

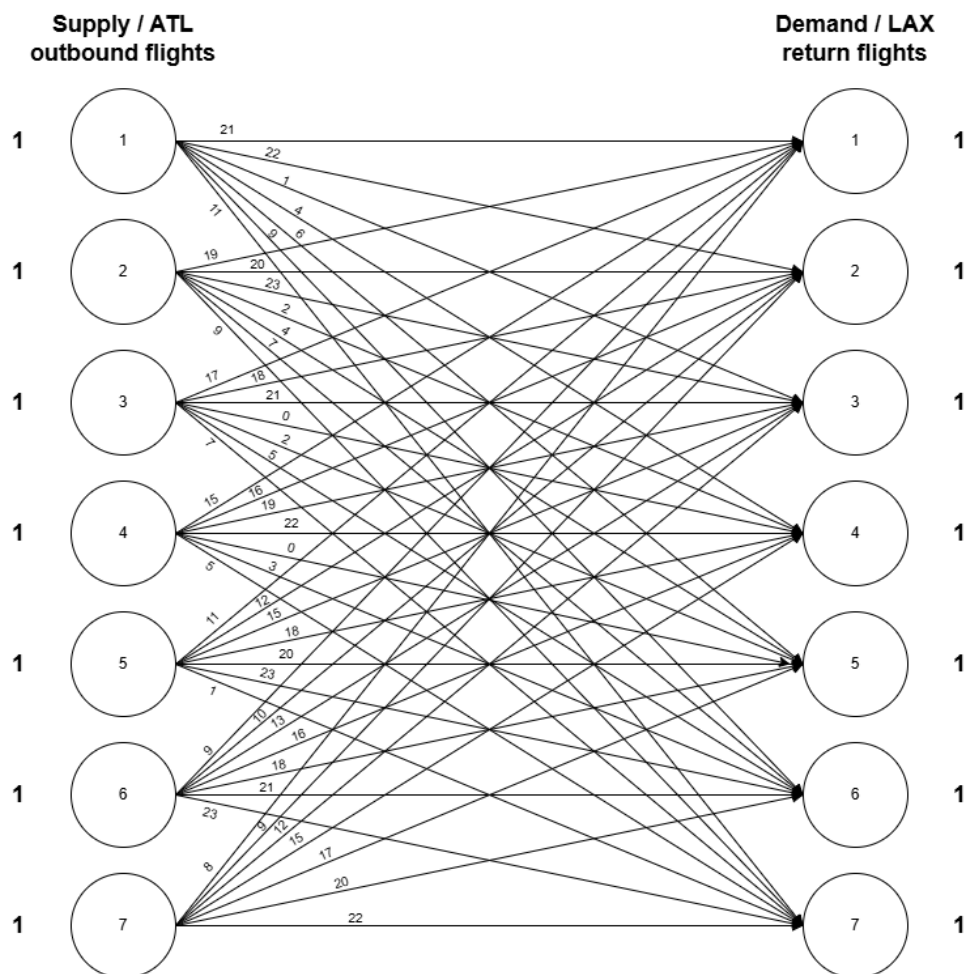
Table 1.1 – Schedule for flights

Flight #	Departs ATL	Arrives LAX	Flight #	Departs LAX	Arrives ATL
A1	6:00 AM	8:00 AM	L1	5:00 AM	9:00 AM
A2	8:00 AM	10:00 AM	L2	6:00 AM	10:00 AM
A3	10:00 AM	12:00 PM	L3	9:00 AM	1:00 PM
A4	12:00 PM	2:00 PM	L4	12:00 PM	4:00 PM
A5	4:00 PM	6:00 PM	L5	2:00 PM	6:00 PM
A6	6:00 PM	8:00 PM	L6	5:00 PM	9:00 PM
A7	7:00 PM	9:00 PM	L7	7:00 PM	11:00 PM

Source: initial data

The network flow model that represents the problem is shown below.

Figure 1.1 – Network flow model for Omega Airline flights



Source: author's development

The model is balanced, because there are 7 supply nodes (each with value 1) and 7 demand nodes (each with value 1 as well). Supply nodes are represented as ATL-LAX

(LA airport) flights. On the other hand, demand nodes are shown as LAX-ATL flights. Arcs represent feasible crew connections that satisfy the FAA one-hour rest rule. Arc costs represent the layover time, which are seen on arcs in Figure 1.1 or table below.

Table 1.2 – Arc costs (layover time) between each flight possible

Flights	L1	L2	L3	L4	L5	L6	L7
A1	21	22	1	4	6	9	11
A2	19	20	23	2	4	7	9
A3	17	18	21	24	2	5	7
A4	15	16	19	22	24	3	5
A5	11	12	15	18	20	23	1
A6	9	10	13	16	18	21	23
A7	8	9	12	15	17	20	22

Source: author's development

2. Mathematical Formulation

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ be the set of ATL-LAX outbound flights, indexed by i and $L = \{1, 2, 3, 4, 5, 6, 7\}$ be the set of LAX-ATL return flights, indexed by j .

Let c_{ij} be the layover cost, which is measured in hours, if the crew from ATL-LAX flight is assigned to LAX-ATL return flight j . The value of c_{ij} is calculated as the waiting time between the arrival of flight i in LAX and the departure of return flight j to ATL. If the return flight has already departed on the same day, the next-day departure is used. Let $s_i = 1$ for each $i \in A$ be the supply of each ATL-LAX flight and $d_j = 1$ for each $j \in L$ be the demand of each LAX-ATL return flight. Therefore, the model is balanced because total supply and demand are equal.

Decision variables are the following:

$$x_{ij} = \begin{cases} 1, & \text{if flight } i \text{ from ATL is assigned to flight } j \text{ from LAX} \\ 0, & \text{otherwise} \end{cases}$$

For instance, if $x_{13} = 1$, then the crew of A1 flight take L3 return flight.

Objective function is to determine an assignment of crews to flights that minimizes the total waiting time for Omega Airlines. Thus, the minimisation function will look like this:

$$\begin{aligned} \text{Min} \quad & 21x_{11} + 22x_{12} + x_{13} + 4x_{14} + 6x_{15} + 9x_{16} + 11x_{17} + 19x_{21} + 20x_{22} + 23x_{23} \\ & + 2x_{24} + 4x_{25} + 7x_{26} + 9x_{27} + 17x_{31} + 18x_{32} + 21x_{33} + 24x_{34} + 2x_{35} \\ & + 5x_{36} + 7x_{37} + 15x_{41} + 16x_{42} + 19x_{43} + 22x_{44} + 24x_{45} + 3x_{46} + 5x_{47} \\ & + 11x_{51} + 12x_{52} + 15x_{53} + 18x_{54} + 20x_{55} + 23x_{56} + x_{57} + 9x_{61} + 10x_{62} \\ & + 13x_{63} + 16x_{64} + 18x_{65} + 21x_{66} + 23x_{67} + 8x_{71} + 9x_{72} + 12x_{73} \\ & + 15x_{74} + 17x_{75} + 20x_{76} + 22x_{77} \end{aligned}$$

Or:

$$\text{Min } Z = \sum_{i \in A} \sum_{j \in L} c_{ij} x_{ij}$$

Where c_{ij} is the layover time between ATL-LAX flight i and LAX-ATL return flight j .

There are two different categories of constraints for the following task. Each flight from ATL should get exactly one return flight:

$$\begin{aligned}
x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} &= 1 \\
x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} &= 1 \\
x_{31} + x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} &= 1 \\
x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} &= 1 \\
x_{51} + x_{52} + x_{53} + x_{54} + x_{55} + x_{56} + x_{57} &= 1 \\
x_{61} + x_{62} + x_{63} + x_{64} + x_{65} + x_{66} + x_{67} &= 1 \\
x_{71} + x_{72} + x_{73} + x_{74} + x_{75} + x_{76} + x_{77} &= 1
\end{aligned}$$

And each flight from LAX should be used only once:

$$\begin{aligned}
x_{11} + x_{21} + x_{31} + x_{41} + x_{51} + x_{61} + x_{71} &= 1 \\
x_{12} + x_{22} + x_{32} + x_{42} + x_{52} + x_{62} + x_{72} &= 1 \\
x_{13} + x_{23} + x_{33} + x_{43} + x_{53} + x_{63} + x_{73} &= 1 \\
x_{14} + x_{24} + x_{34} + x_{44} + x_{54} + x_{64} + x_{74} &= 1 \\
x_{15} + x_{25} + x_{35} + x_{45} + x_{55} + x_{65} + x_{75} &= 1 \\
x_{16} + x_{26} + x_{36} + x_{46} + x_{56} + x_{66} + x_{76} &= 1 \\
x_{17} + x_{27} + x_{37} + x_{47} + x_{57} + x_{67} + x_{77} &= 1
\end{aligned}$$

Because it is an assignment task, the job is either performed or not, which is a binary condition:

$$x_{ij} \in \{0,1\} \quad \forall i \in A, \forall j \in L$$

3. Model Implementation

The mathematical model was implemented in IBM ILOG CPLEX Optimisation Studio using OPL. The .mod file defines the sets of ATL-LAX outbound flights and LAX-ATL return flights, the binary decision variable x_{ij} , the objective function and the assignment constraints. The .dat file imports the layover cost matrix from the Excel file using SheetRead, where the numerical values from Table 1.2 are stored. After solving the model, CPLEX returned an optimal objective value of 27 hours. The selected assignments are shown in Table 1.3, while all other decision variables have a value of 0.

Table 1.3 – Optimal crew assignment obtained from CPLEX

Chosen ATL-LAX Flight #	Chosen LAX-ATL Flight #	Layover time, cost
A1	L3	1
A2	L4	2
A3	L5	2
A4	L6	3
A5	L7	1
A6	L2	10
A7	L1	8
Optimal layover time (h)		27

Source: author's analysis

4. Results and discussion

The problem considers the optimal assignment of Omega Airlines crews from ATL-LAX outbound flights to LAX-ATL return flights. The objective was to minimise the total crew layover time, while ensuring that every outbound crew is assigned to exactly one return flight and every return flight is covered by one crew. The model was solved and the optimal objective value was 27 hours.

The optimal assignment is as follows: A1 is assigned to L3 with layover of 1 hour, A2 to L4 with 2 hours, A3 to L5 with 2 hours, A4 to L6 with 3 hours, A5 to L7 with 1 hour, A6 to L2 with 10 hours and A7 to L1 with 8 hours. This gives a total layover time of 27 hours.

The solution satisfies the FAA rest requirements because every selected assignment gives the crew at least one hour of rest before the return flight departs. The first five crews are assigned to same-day return flights with short waiting times, while the two late arrivals are assigned to next-day morning flights. Although overnight waiting is longer, it is unavoidable because no later same day return flights are available. Therefore, the model produces an efficient schedule by selecting the shortest feasible layovers and avoiding unnecessary long waiting times.

Question 2: COVID-19 Healthcare Facility Capacity Optimization Problem

1. Mathematical formulation

CDCP must allocate COVID-19 patients from 9 counties to existing and potential temporary healthcare facilities. The model decides patient flows, temporary facility construction and emergency extra capacity. It minimises transportation, construction and penalty costs while satisfying all demand and capacity restrictions.

Let $I = \{1, 2, \dots, 9\}$ be the set of counties, $E = \{1, 2, \dots, 14\}$ be the set of existing healthcare facilities, $T = \{15, \dots, 23\}$ be the set of potential temporary facilities and $J = E \cup T$ be the set of all healthcare facilities.

Let d_i be the demand for COVID-19 hospital treatment in county i , cap_j be the capacity of facility j , F_j be the fixed cost of building temporary facility j , in this problem: $F_j = 500\,000$, $j \in T$. Let (X_i, Y_i) be the coordinates of county i , (A_j, B_j) be the coordinates of facility j . The distance between county i and facility j is:

$$dist_{ij} = \sqrt{(X_i - A_j)^2 + (Y_i - B_j)^2}$$

For sensitivity analysis, demand is updated as $d'_i = \text{round}(1.2d_i)$, $\forall i \in I$, where Rounding d_i is the original demand and d'_i is the rounded demand used in the 20% increase scenario.

Euclidean distance was used because county and facility locations are given as coordinate points. The transportation cost per patient from county i to facility j is:

$$c_{ij} = 5 * dist_{ij}$$

because one coordinate unit represents 10 miles and each 10-mile increase costs 5 dollars per patient.

Decision variables:

Let $x_{ij} \geq 0$ be the number of COVID-19 patients from county i allocated to healthcare facility j .

Let:

$$y_j = \begin{cases} 1, & \text{if temporary facility } j \text{ is built} \\ 0, & \text{otherwise} \end{cases}$$

where $j \in T$.

Extra capacity is also mentioned, so $z_j \geq 0$ – amount of extra capacity added at temporary facility j , if needed.

Let P be the penalty cost for one unit of emergency extra capacity. In this model, $P = 1\,000\,000$ dollars.

Objective function is a minimisation function of total system cost:

$$\min Z = \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{j \in T} F_j y_j + P * \sum_{j \in T} z_j$$

where first part is total transportation cost, second part is total temporary facility construction cost, and third part is a penalty cost for adding extra capacity. Since P is very large (1 000 000 in this case), extra capacity is used only when ordinary capacity is insufficient.

There are 3 main constraints for this problem. The first one to satisfy demand of every county:

$$\sum_{j \in J} x_{ij} = d_i, \forall i \in I$$

which means all patients from every county must be assigned to some healthcare facility.

Second constraint – existing hospitals cannot receive more patients than their capacity:

$$\sum_{i \in I} x_{ij} \leq cap_j, \forall j \in E$$

Third constraint – the capacity of a temporary facility cannot be exceeded. A temporary facility can receive patients only if it is built. Emergency extra capacity can increase the available capacity, but it receives a penalty cost in the objective function:

$$\sum_{i \in I} x_{ij} \leq cap_j * y_j + z_j, \forall j \in T$$

Extra capacity can only be added at a temporary facility if that facility is built:

$$z_j \leq M y_j, \quad \forall j \in T$$

Where $M = y_j \sum_{i \in I} d_i$ is a sufficiently large number equal to total demand.

The binary construction decision constraint is:

$$y_j \in \{0,1\}, \quad \forall j \in T$$

The integer non-negativity constraints are:

$$x_{ij} \geq 0, \quad \forall j \in J \text{ and } \forall i \in I$$

$$z_i \geq 0, \quad \forall j \in T$$

2&3. Coding and presenting results

The OPL code is in the Appendix and uses separate .mod and .dat files. The baseline optimal objective value is 1 521 653.42 dollars: 21 653.42 transportation cost, 1 500 000 construction cost and 0 penalty cost.

The baseline solution builds facilities 15, 17, 18. Each adds 100 places, so 300 temporary places are created and the construction budget used in 1 500 000 dollars.

One optimal allocation of patients is as follows:

Table 2.1 – Optimal allocation of patients (baseline scenario)

County	Facility	Patients allocated
County 1	Facility 1	281
	Facility 15	57
	Facility 18	13
County 2	Facility 2	187
	Facility 15	43
County 3	Facility 3	200
	Facility 4	223
	Facility 7	6
	Facility 17	100
County 4	Facility 5	281
	Facility 18	58
County 5	Facility 6	281
	Facility 7	50
	Facility 18	29
County 6	Facility 7	166
	Facility 8	200
	Facility 11	161
County 7	Facility 9	250
	Facility 10	125
	Facility 13	94
County 8	Facility 12	28
	Facility 13	206
County 9	Facility 12	272
	Facility 14	228

Source: author's analysis

Total demand is 3539 patients, and all 3539 are served, so demand served equals 100%. No emergency extra capacity is required.

4. Sensitivity analysis

In the sensitivity scenario, demand is increased by 20% and rounded, so total demand rises from 3539 to 4247 patients. The new optimal objective value is 71 525 513.31 dollars: 25 513.31 transportation cost, 4 500 000 construction cost and 67 000 000 penalty cost. Demand served remains 100%.

The higher-demand solution builds all nine temporary facilities: 15-23. Construction cost rises by 3 000 000 dollars. Emergency extra capacity is required at Facility 15 (40 patients) and Facility 17 (27 patients).

The allocation changes because three temporary facilities are no longer enough.

Table 2.2 Optimal patient allocation under the 20% demand increase scenario

County	Facility	Patients allocated
1	Facility 1	281
	Facility 15	140
2	Facility 2	187
	Facility 16	89
3	Facility 3	200
	Facility 4	223
	Facility 7	74
	Facility 16	11
	Facility 17	127
4	Facility 5	281
	Facility 18	100
	Facility 21	26
5	Facility 6	281
	Facility 19	100
	Facility 20	51
6	Facility 7	148
	Facility 8	200
	Facility 11	187
	Facility 20	49
	Facility 23	48
7	Facility 9	250
	Facility 10	125
	Facility 13	188
8	Facility 13	55
	Facility 21	74
	Facility 22	100
	Facility 23	52
9	Facility 12	300
	Facility 13	57

Source: author's analysis

Existing facilities are used at or close to capacity, so extra patients are redirected to temporary locations. The penalty cost is high because 67 emergency places are used at 1 000 000 dollars each.

Table 2.3 – Comparison of baseline and 20% demand increase scenarios

Indicator	Baseline	Sensitivity	Interpretation
Demand served	3539 (100%)	4247 (100%)	All demand served
Temporary facilities	3: 15, 17, 18	9: 15-23	All sites needed
Build cost	1 500 000	4 500 000	+3 000 000
Transport cost	21 653.42	25 513.31	Small increase
Extra capacity	0	67: F15 (40), F17 (27)	Penalty triggered
Penalty cost	0	67 000 000	Main cost driver
Total cost	1 521 653.42	71 525 513.31	Much higher

Source: author's analysis

The transportation cost increases only slightly because the model can still assign many patients to nearby facilities. The main cost increase comes from capacity shortage: all temporary facilities must be opened, and 67 additional emergency places are required. This shows that the system is much more sensitive to capacity limits than to travel distance. Overall, the system remains feasible under the 20% demand increase because 100% of patients are still served. However, the solution becomes much more expensive and capacity-constrained, mainly due to the construction of all temporary facilities and the use of emergency extra capacity.

Appendix

Question 1: Optimally Assigning Crews to Flights

Code for .mod file

```
range A = 1..7;
range L = 1..7;

int c[A][L] = ...;

// decision variable
dvar boolean x[A][L];

// objective function
minimize sum(i in A, j in L) c[i][j] * x[i][j];

// constraints
subject to{
    forall(i in A)
        sum(j in L) x[i][j] == 1;

    forall(j in L)
        sum(i in A) x[i][j] == 1;
}
```

Source: author's development

Code for .dat file

```
SheetConnection sheet("Excel Task 1.xlsx");
c from SheetRead(sheet, "Sheet1!C16:I22");
```

Source: author's development

Question 2: COVID-19 Healthcare Facility Capacity Optimization Problem

Code for .mod file

```
// sets
range Counties = 1..9;
range Facilities = 1..23;
range Existing = 1..14;
range Temp = 15..23;

// data from .dat
int baseDemand[Counties] = ...;
float demandMultiplier = ...;

float countyX[Counties] = ...;
float countyY[Counties] = ...;

float facilityX[Facilities] = ...;
float facilityY[Facilities] = ...;

int capacity[Facilities] = ...;

float costPerDistanceUnit = ...;
float fixedBuildCost = ...;
```

```

int penaltyCost = ...;

// demand
float demand[i in Counties] = round(baseDemand[i] * demandMultiplier);

// total demand, used as Big M
float totalDemand = sum(i in Counties) demand[i];

// distance and transportation cost
float distance[i in Counties][j in Facilities] =
    sqrt((countyX[i] - facilityX[j]) * (countyX[i] - facilityX[j])
    + (countyY[i] - facilityY[j]) * (countyY[i] - facilityY[j]));

float transportCost[i in Counties][j in Facilities] =
    costPerDistanceUnit * distance[i][j];

// decision variables
dvar int+ x[Counties][Facilities]; // patients from county i to facility j
dvar boolean build[Temp]; // 1 if temporary facility j is built
dvar int+ extra[Temp]; // extra capacity added at temporary facility j

// objective parts
dexpr float totalTransportCost =
    sum(i in Counties, j in Facilities) transportCost[i][j] * x[i][j];

dexpr float totalBuildCost =
    sum(j in Temp) fixedBuildCost * build[j];

dexpr float totalPenaltyCost =
    penaltyCost * sum(j in Temp) extra[j];

dexpr int totalServed =
    sum(i in Counties, j in Facilities) x[i][j];

dexpr float percentageServed =
    100 * totalServed / totalDemand;

// objective function
minimize totalTransportCost + totalBuildCost + totalPenaltyCost;

// constraints
subject to {

    // each county demand must be fully satisfied
    forall(i in Counties)
        sum(j in Facilities) x[i][j] == demand[i];

    // existing facility capacity cannot be exceeded
    forall(j in Existing)
        sum(i in Counties) x[i][j] <= capacity[j];

    // temporary facility capacity can be used only if facility is built
    forall(j in Temp)
        sum(i in Counties) x[i][j] <= capacity[j] * build[j] + extra[j];

    // extra capacity can only be added if the temporary facility is built
    forall(j in Temp)
        extra[j] <= totalDemand * build[j];
}

```

Source: author's development

Code for .dat file

```
// original case: use 1.0
// sensitivity case: change this to 1.2
demandMultiplier = 1.0;

// cost settings
costPerDistanceUnit = 5;
fixedBuildCost = 500000;
penaltyCost = 1000000;

// county demand
baseDemand = [351, 230, 529, 339, 360, 527, 469, 234, 500];

// county coordinates
countyX = [1, 3, 5.5, 1, 3, 5.5, 1, 3, 4.5];

countyY = [1.5, 1, 1.5, 4.5, 3.5, 4.5, 8, 6, 8];

// facility coordinates
facilityX = [1, 2.5, 5, 6.5, 1, 3, 5, 6.5, 1, 1.5, 8.5, 5, 3,
            6, 1.5, 3.5, 5.5, 1.5, 3.5, 4.5, 1.5, 3.5, 5.5
];

facilityY = [2, 1, 1, 3.5, 5, 4, 4, 5.5, 8.5, 9.5, 6, 8, 9,
            9, 1, 1.5, 2.5, 3.5, 2.5, 4.5, 6.5, 6.5, 6.5
];

// facility capacities
capacity = [281, 187, 200, 223, 281, 281, 222, 200, 250, 125, 187,
            300, 300, 243, 100, 100, 100, 100, 100, 100, 100, 100, 100
];
```

Source: author's development